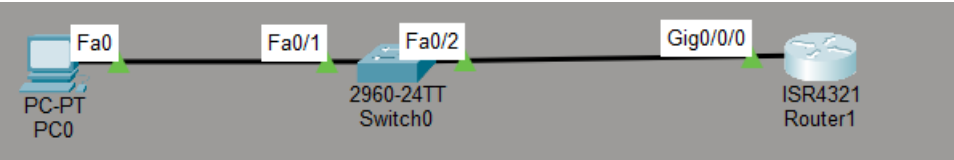


Q-Simulate a remote access VPN scenario in Packet Tracer by configuring a VPN client PC to connect securely to one of the branch routers using the built-in VPN client feature.
Hint: Use the VPN tab on the PC and set the correct server IP, username, and password.

To simulate a **Remote Access VPN** scenario in Cisco Packet Tracer by configuring a VPN client PC to securely connect to a branch router using the built-in **VPN Client** feature.



IP Addressing

Device	Interface	IP Address	Subnet Mask
Router0	G0/0/0	192.168.10.1	255.255.255.0
PC0	FastEthernet0	192.168.10.10	255.255.255.0

➤ **Router Interface Configuration**

```
enable
configure terminal
interface GigabitEthernet 0/0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
```

➤ **VPN User Configuration**

```
username vpnuser password cisco123
aaa new-model
aaa authentication login VPN_AUTH local
aaa authorization network VPN_GROUP local
```

➤ **VPN Group Configuration**

```
crypto isakmp client configuration group VPNGROUP  
key VPNKEY  
pool VPNPOOL  
exit  
ip local pool VPNPOOL 192.168.20.10 192.168.20.20  
access-list 100 permit ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
```

➤ **ISAKMP Configuration**

```
crypto isakmp policy 10  
encryption aes  
hash sha  
authentication pre-share  
group 2  
exit  
crypto isakmp key VPNKEY address 0.0.0.0
```

➤ **IPsec Configuration**

```
crypto ipsec transform-set VPN_SET esp-aes esp-sha-hmac  
crypto dynamic-map VPN_DYNAMIC 10  
set transform-set VPN_SET  
exit  
crypto map VPN_MAP 10 ipsec-isakmp dynamic VPN_DYNAMIC  
interface GigabitEthernet 0/0/0  
crypto map VPN_MAP
```

exit

crypto map VPN_MAP client authentication list VPN_AUTH

crypto map VPN_MAP isakmp authorization list VPN_GROUP

crypto map VPN_MAP client configuration address respond

➤ **VPN Client Configuration On Pc0**

VPN Parameter	Value
Group Name	VPNGROUP
Group Key	VPNKEY
Host IP (Server IP)	192.168.10.1
Username	vpnuser
Password	cisco123

➤ **Verification**

show crypto map

The basic network connectivity was successfully verified. The VPN connection itself requires a more realistic topology with the VPN client located on a **separate remote/WAN network** from the branch VPN server.